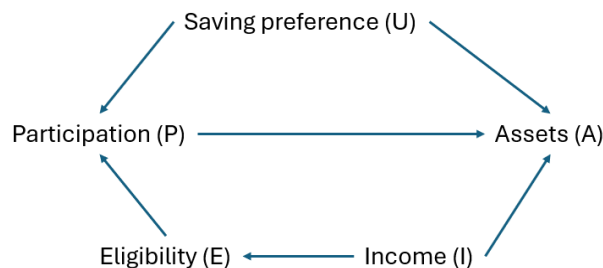


Project: Causal Machine Learning

To estimate the causal effect of 401(k) plans on accumulated assets, we will focus on eligibility rather than participation, since the latter can be influenced by saving preferences. In contrast, eligibility can be seen as exchangeable conditional on income and the causal effect of eligibility can thus be directly estimated.

1. To explain why eligibility for a 401(k) is indeed exchangeable conditional on income, we can consider the following causal diagram.



This causal diagram shows that income is a confounder on the (indirect) effect of eligibility on accumulated assets. Conditioning on this confounder means that the backdoor paths of $I \rightarrow E \rightarrow P$ and $I \rightarrow A$ will be blocked. Then, within each strata of income, people eligible for a 401(k) are statistically similar to those who are not eligible. In our case: $E \perp\!\!\!\perp A_0 | I$ and $E \perp\!\!\!\perp A_1 | I$. This makes it possible to assess the indirect effect of 401k eligibility on accumulated assets through participation.

2. We are interested in understanding the effect of eligibility across the whole population, as opposed to only focusing on the subgroup receiving treatment. Therefore, we choose to report the average treatment effect (ATE) to estimate the effect of eligibility for enrolling in a 401(k) plan on net financial assets, controlling for potential confounders. The ATE then reports the expected differences in assets if everyone were eligible for a 401(k) as opposed to if no one was eligible.

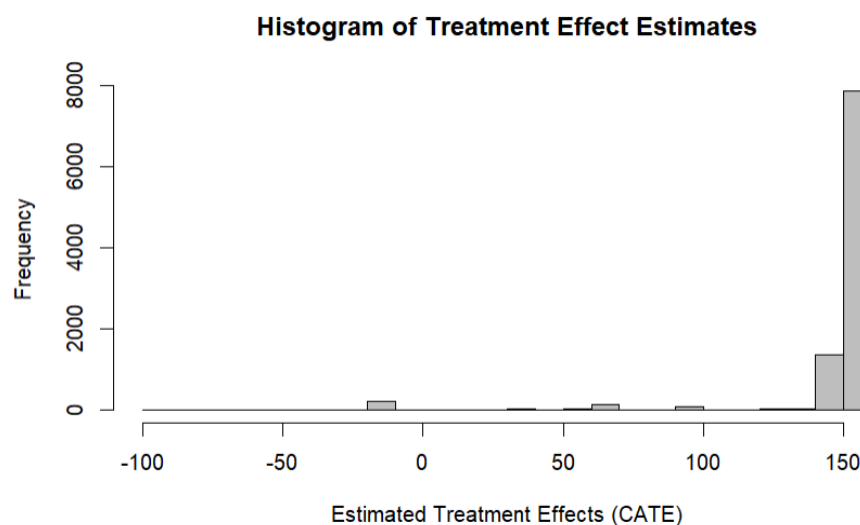
To get an accurate estimate of the ATE, we first calculate the augmented inverse probability weights (AIPW), using 5-fold cross-fitting to reduce bias from overfitting. After running the code with the use of SuperLearner, we get the following estimate: $E\{Y(1)-Y(0)\} = 2705.6$ with a 95% confidence interval of [137.59, 5273.69]. This means that individuals eligible for a 401(k) have significantly higher net financial assets (+\$2,705.6) than individuals who are not eligible. We additionally check the inverse probability weights (IPW) and influence functions (IF) to check the stability and robustness of our estimator. The mean IPW is 1.68, indicating that individuals eligible for a 401(k) might be underrepresented and therefore receive a bigger weight. Considering 6% of observations have large weights (>5), this indicates some instability in the estimate. The IFs show a range of -3.7M to +9.6M with a median of -3583, indicating a right-tailed distribution. This means that some observations have a big influence on the estimate, showing that the estimate is not very robust.

Similarly, we calculate the targeted maximum likelihood estimator (TMLE) with SuperLearner, using 5-fold cross fitting. TMLE resulted in an ATE estimate of -507.02 with a 95% confidence interval of [-2985.58, 1971.54]. As the confidence interval contains 0, there is no evidence for a difference in net financial assets between individuals eligible for a 401(k) and individuals who are not eligible. The mean IPW is 1.71 and the influence curves show a range of -3.9M to +9.2M with a median of -1808, once again showing that the estimate is relatively unstable and not adequately robust. However, the abovementioned double robust estimators clearly show the importance of taking into account covariates in the model. When estimating the ATE without adjusting for covariates, this reveals an ATE of 19559. This would mean that individuals eligible for a 401(k) have significantly higher net financial assets (+\$19,559.34) than individuals who are not eligible. This clearly over-estimates the causal effect of eligibility and would lead to false conclusions.

3. As the previous estimates were not ideal, we now conduct a more powerful test by testing $E\{\text{Cov}(A, Y | L)\} = 0$. The SuperLearner was once again used to estimate $E(A|W)$ and $E(Y|W)$. After calculating the efficient influence function based on the residuals, we obtained the following estimate: $\theta = -382.10$ (SE = 167.14). The p-value is 0.02, indicating significant evidence for a conditional association between eligibility and net financial assets after adjusting for covariates, where being eligible for a 401(k) is typically associated with lower assets.

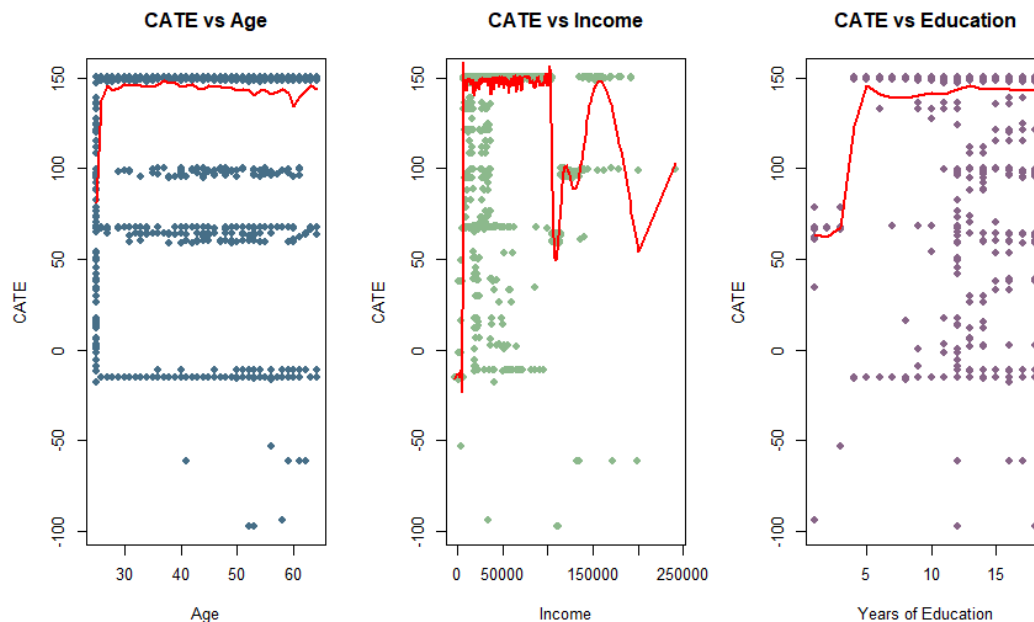
4. Next, the heterogeneity in the effect of 401(k) eligibility on net financial assets was explored. We implemented the R-learner framework with 5-fold cross-fitting to mitigate overfitting bias. This method provides debiased estimates of the conditional average treatment effect (CATE), while controlling for certain covariates. We modeled the treatment effect in function of age, income, and years of education, adjusting for the relevant covariates from question 2. The R-learner was implemented using the *rboost* function from the R package *rlearner*, a gradient boosting algorithm.

The histogram of the estimated conditional average treatment effects (CATE) shows a pronounced right-skewed distribution with most individuals having positive treatment effects clustered near 150. This indicates that for many people, eligibility for a 401(k) plan is associated with an increase of about \$150 in net financial assets compared to those not eligible, controlling for covariates. However, there is a long left tail extending to smaller positive or even negative values, suggesting that some individuals experience little to no benefit, or even a negative association, from eligibility. This heterogeneity underscores the importance of examining individual-level effects rather than relying solely on average treatment effects. The large heterogeneity also becomes clear in the next figure (see next paragraph), where groups with very different CATEs become clear.



The resulting CATE estimates for each individual were then visualized for age, income and education separately (see figure below). The red line in each plot shows a smoothed estimate of how the treatment effect varies by the covariate of interest, controlling for all relevant variables. For age, the treatment effect remains relatively stable between ages 35 and 60, with a modest peak around age 35, suggesting that 401(k) eligibility has the greatest impact during prime working years when individuals are most likely to save and accumulate assets. However, a dip is observed at younger ages (below 30), which may reflect that younger individuals are less likely to participate or fully benefit from the plan possibly due to limited work experience, lower savings capacity, or less financial awareness. Next, when looking at income, the effect is strongest at low to moderate income levels but drops sharply beyond approximately \$100,000. This decline at higher incomes could indicate that wealthier individuals diversify their investments outside 401(k) plans, thereby reducing the marginal benefit of eligibility. The dip at the very low-income end might reflect financial constraints limiting plan participation or saving behavior, dampening the effect despite eligibility. Finally for education, the

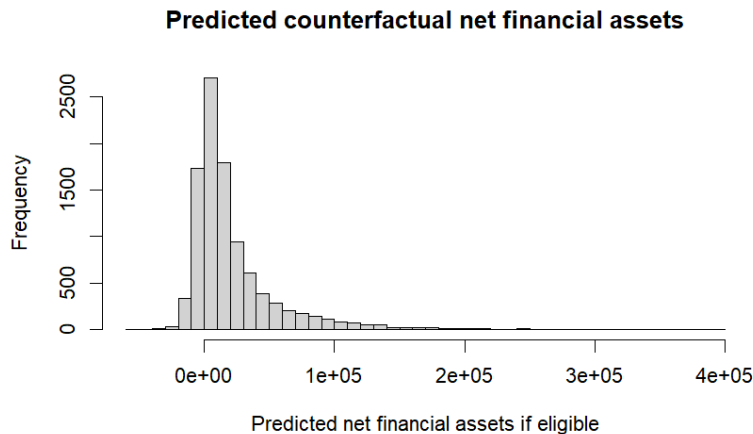
treatment effect generally increases with more years of schooling, leveling off after around 12 years. This pattern likely reflects that individuals with higher education have greater financial literacy and are more likely to participate in and benefit from 401(k) plans. The sudden dip at the lowest education levels may be due to smaller sample sizes or lower financial knowledge hindering the ability to capitalize on 401(k) eligibility.



Finally, to assess estimator stability and reliability, we again examined the IPWs and influence functions. The distribution of control IPWs is well-behaved, with values mostly between 1 and 5, indicating low influence of any single control unit. However, the treated weights have a long right tail, with values exceeding 40, due to low propensity scores ($\hat{p} < 0.05$ for 162 units). These high weights signal potential instability, as a few treated individuals may disproportionately affect estimates. Additionally, we identified 135 extreme pseudo-outcomes ($\pm 3SD$ from the mean). These diagnostics underscore the need to interpret treatment effect heterogeneity with care, particularly in the treated group where the causal estimates appear to rely on a few, potentially unrepresentative observations.

5. In this part of the project, we aimed to predict the counterfactual net financial assets under 401(k) eligibility for each individual, conditional on their age, income and years of education, while controlling for a broader set of covariates. We used nested cross-validation with 5-fold cross-fitting. First, in the outer loop, we split the data into five parts. Each time, we trained the models on four parts and tested on the fifth. Then, inside the training data, we did another 5-fold split to fit the outcome model and the propensity score model using the SuperLearner. We then used these models to create pseudo-outcomes. After that, we trained a new model on these pseudo-outcomes using the original covariates. This way, we could predict the counterfactual outcomes for everyone.

The histogram below shows the distribution of the predicted net financial assets that individuals would have if they were eligible for a 401(k) plan. The distribution is right-skewed, with most predicted values lying between \$0 and \$50,000 squared assets. The large peak around low positive values suggests that a majority of individuals would not accumulate extremely large assets even if eligible. The long right tail indicates that a small number of individuals are predicted to have very high net financial assets if they were eligible for a 401(k) plan.



To estimate the counterfactual prediction error, just computing and averaging the squared difference between the observed and predicted outcomes among eligible individuals would not give a valid estimate of the marginal mean squared error (MSE). This is because eligible and non-eligible individuals differ in their characteristics. To correct for this, we used AIPW. The estimated MSE for the predicted counterfactual net financial assets is approximately $2.56e^9$ with a 95% confidence interval of $[1.38e^9 ; 3.74e^9]$. The MSE is relatively large, since financial assets can vary widely and have a heavy-tailed distribution. The confidence interval is quite wide as this may reflect the variability in observed assets and the challenges in predicting counterfactual financial outcomes. Since the financial assets have a very wide scale, the DR-learner approach achieved reasonably good predictions.

We again examined the IPWs and influence functions. The histogram of the treated IPWs shows a right-skewed distribution, with most weights concentrated between 1 and 5 and a tail extending up to 26.5. This suggests that, although most treated units have stable weights, a few individuals with low propensity scores result in relatively high weights. This can potentially increase the variance in the estimator. The control IPWs are more tightly concentrated between 1 and 5. This suggests that the weights of the control group are well-behaved, with little indication of instability. The histogram of approximate influence functions is heavily concentrated near zero. Using a ± 3 SD cut-off, we identified 92 extreme influence values. While most individuals had limited impact, this tail indicates that a small number of observations may have a greater influence on the estimator than expected.

Julie Van Wylsberge: question 1-3
 Annabel De Clercq: question 4
 Renée Spoelman: question 5